

CYBERDEFENDERS

BLUE TEAM

ENDPOINT ANALYSIS



HireMeLab_Writeup

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CyberDefenders

Scenario:

Analyze a disk image to investigate multiple forensic cases by examining registry, event logs, and email artifacts using specified forensic tools.

Q1

Solved : 1686

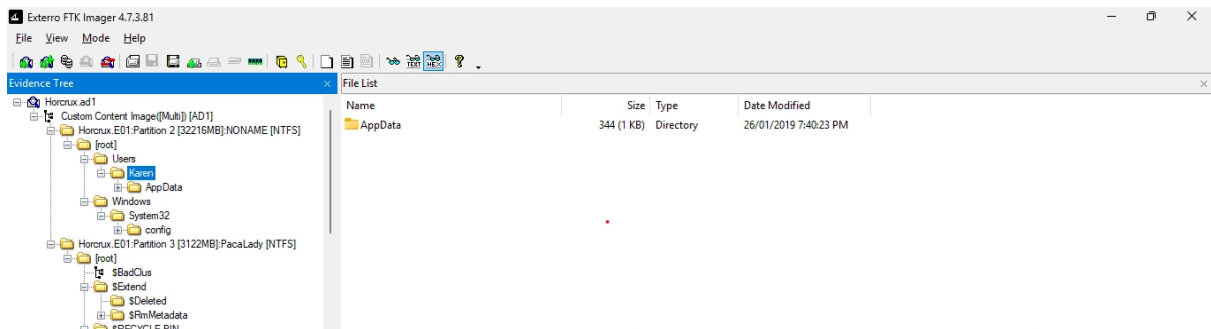
What is the administrator's username?

flag: Karen

process:

Fired up FTK Imager and added the file as an evidence item.

Upon further analysis, it appears that there is only one user in Partition 2 of the `.ad1` file, which is **Karen**.



Q2

Solved : 1542

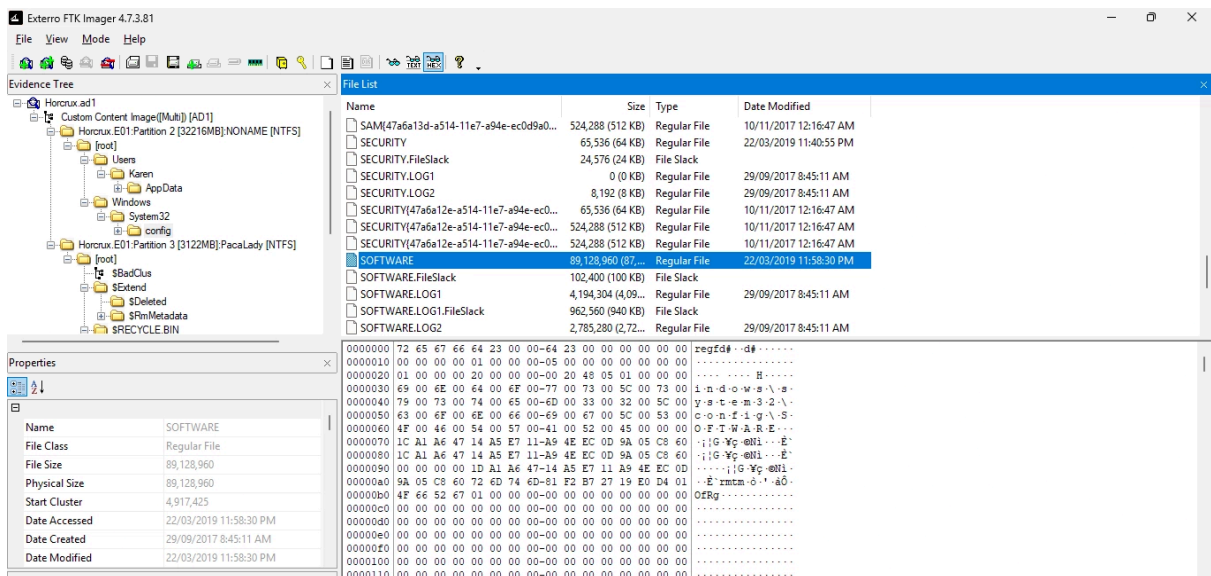
What is the OS's build number?

flag: 16299

process:

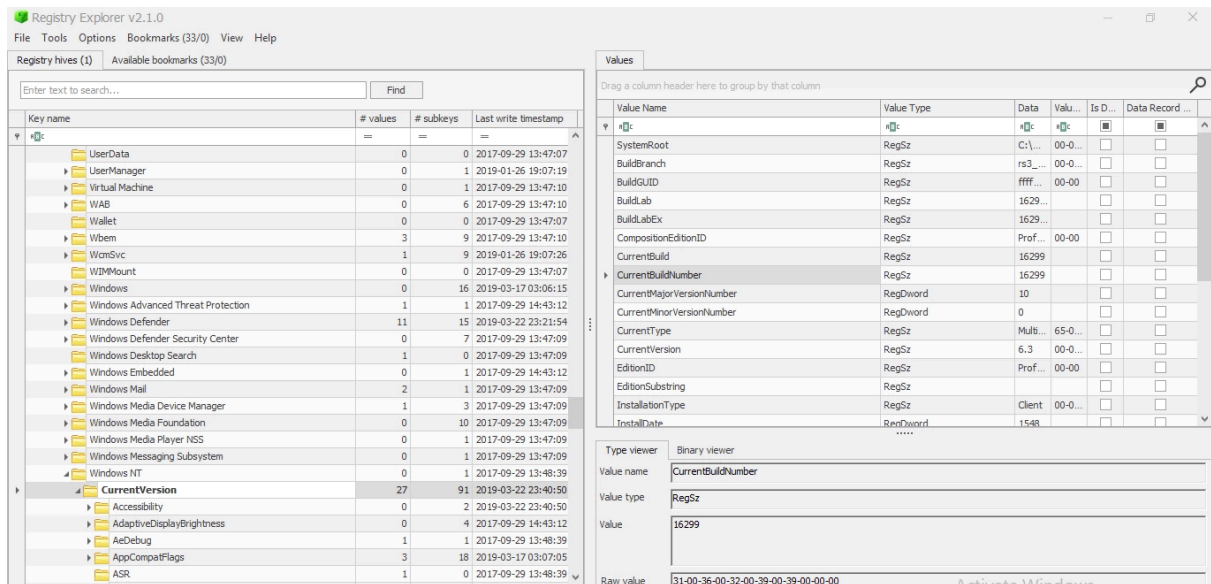
We then proceeded with a quick Google search, which indicated that OS build information is stored in the Windows registry—specifically in the **SOFTWARE** hive located at:

C:\Windows\System32\config\SOFTWARE



I exported the file for further analysis and used Registry Explorer. The hive was loaded to begin the investigation.

From there, we navigated to the **ROOT** folder → **Microsoft** → **Windows NT** → **CurrentVersion**, where the current OS build number can be found.



Q3

Solved : 1522

What is the hostname of the computer?

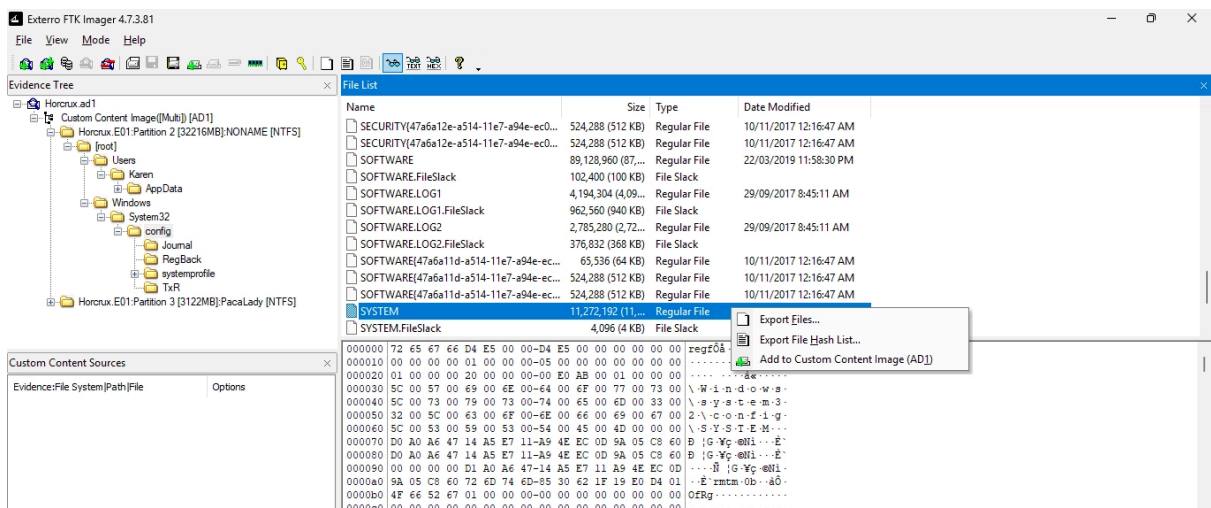
flag: TOTALLYNOTAHACK

process:

We then proceeded with a quick Google search to determine where the hostname is stored in the Windows NTFS file system.

The search results indicated that the hostname can be found in the **SYSTEM** registry hive located at:

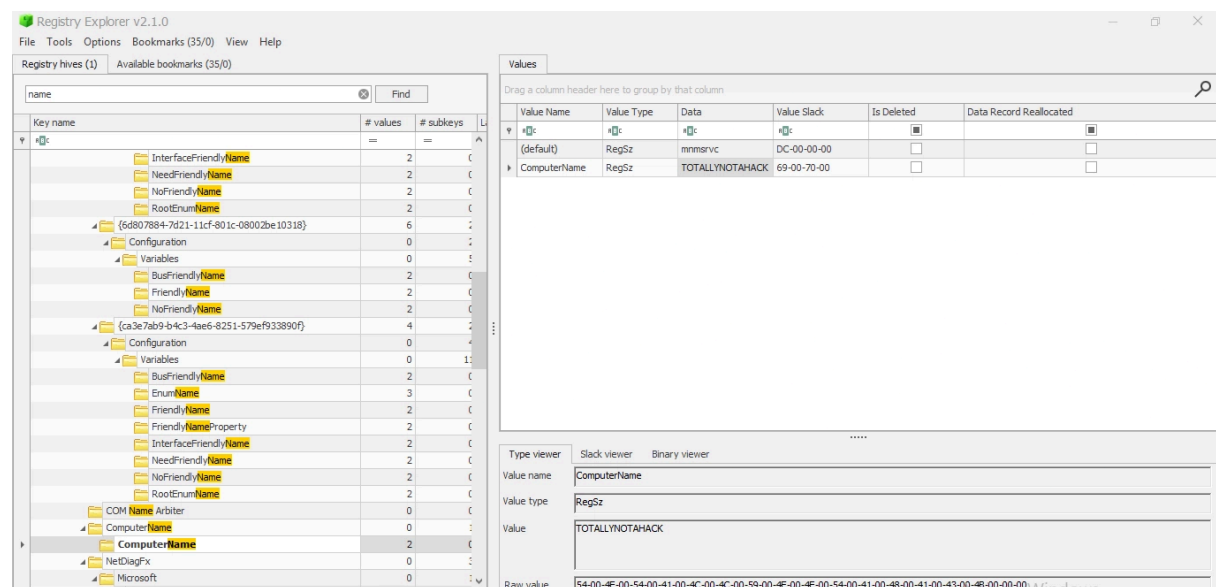
C:\Windows\System32\config\SYSTEM



We then exported the **SYSTEM** file and loaded it into Registry Explorer for further analysis.

Using the filter/search function in Registry Explorer, we searched for relevant keywords such as: **name**, **hostname**, and **computer name**.

From there, we identified that the hostname is located under the **ComputerName** key.



Q4

Solved : 1526

A messaging application was used to communicate with a fellow Alpaca enthusiast. What is the name of the software?

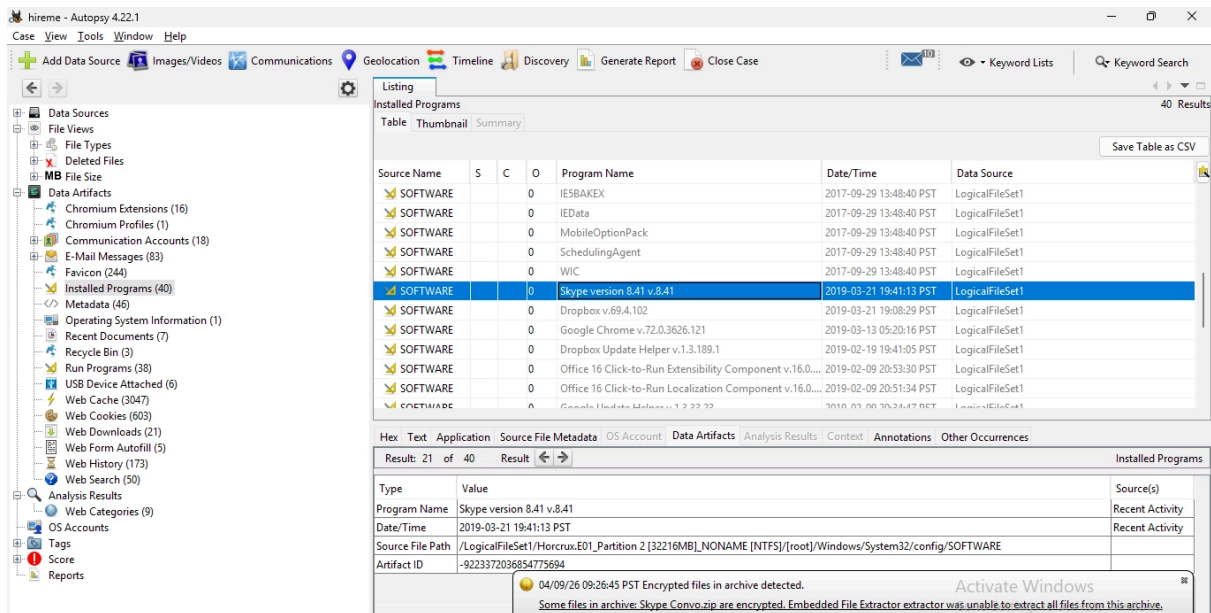
flag: skype

process:

Using FTK Imager, we discovered that a messaging application was used to communicate with a fellow alpaca enthusiast.

This artifact was found in **Partition 3**, specifically within the **Recycle Bin**, where the executable **skype.exe** was identified.

To further validate this finding, a more effective approach is to use **Autopsy**, as it provides a more comprehensive environment for forensic analysis.



Q5

Solved : 1363

What is the zip code of the administrator's post?

flag: 19709

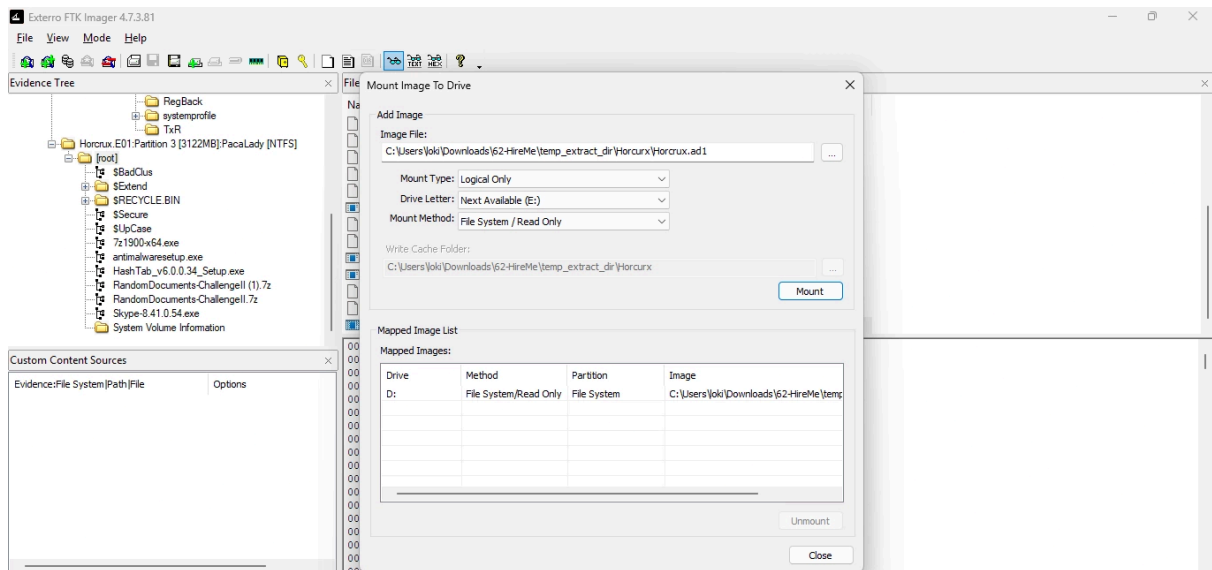
process:

Transition to Autopsy

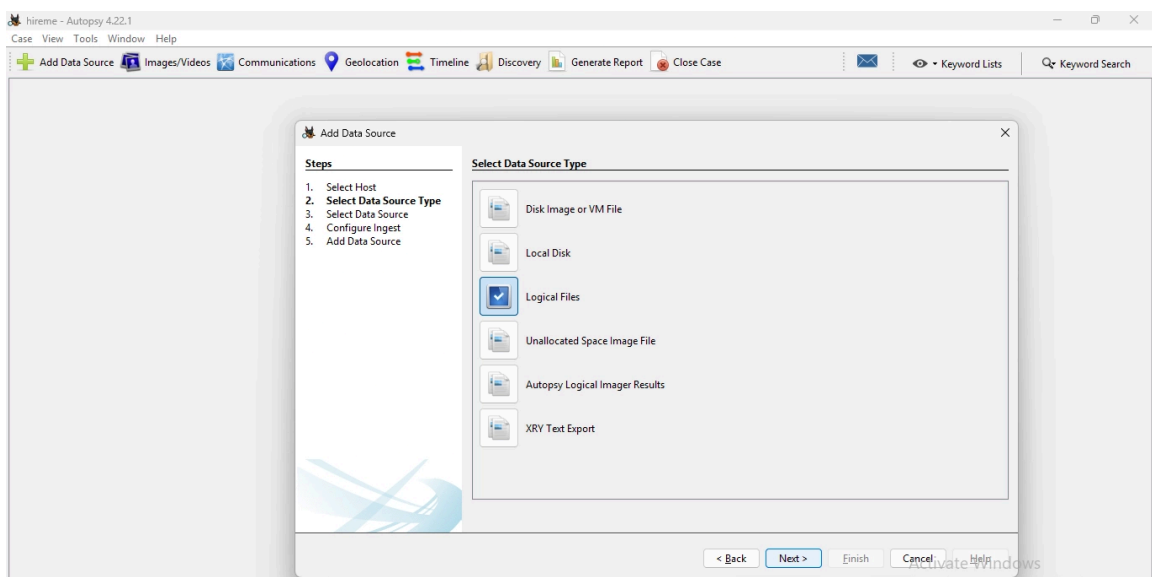
We now proceed to use **Autopsy** for deeper forensic analysis.

Preparing the Data Source

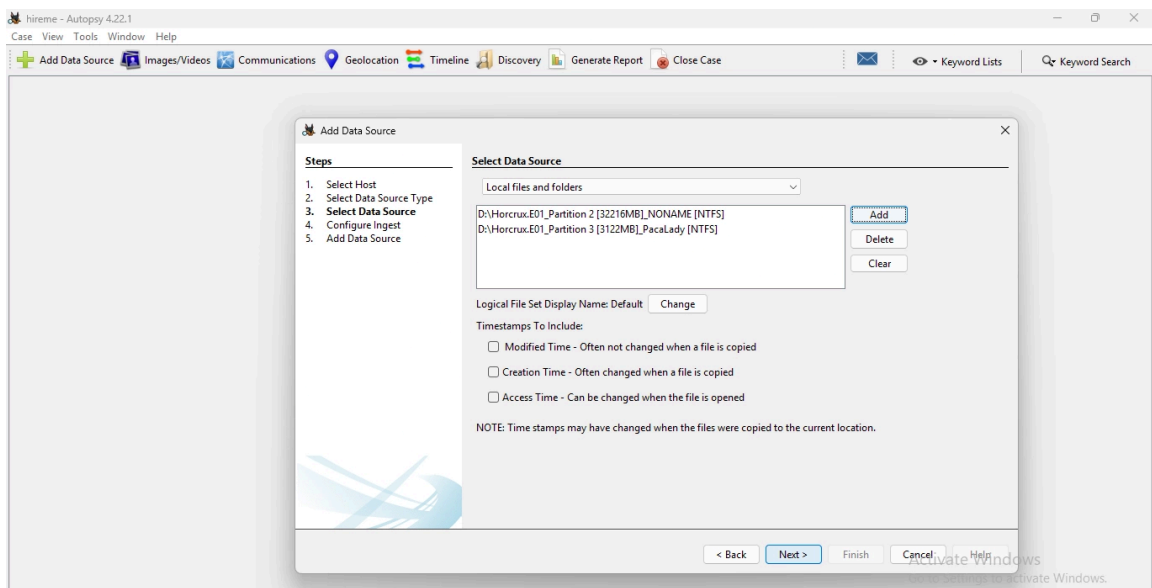
- Mount the **.ad1** file using **FTK Imager**



- or Export the files (**Right-click → Export Files**)
- In **Autopsy**:
 - Go to **Add Data Source**
 - Select **Logical Files**



- Click **Add**, then choose the previously exported files or mounted ad1 partitions



Analysis Approach

After gaining some familiarity with **Registry Explorer**, we move to **Autopsy**, which offers more advanced capabilities for in-depth forensic investigation.

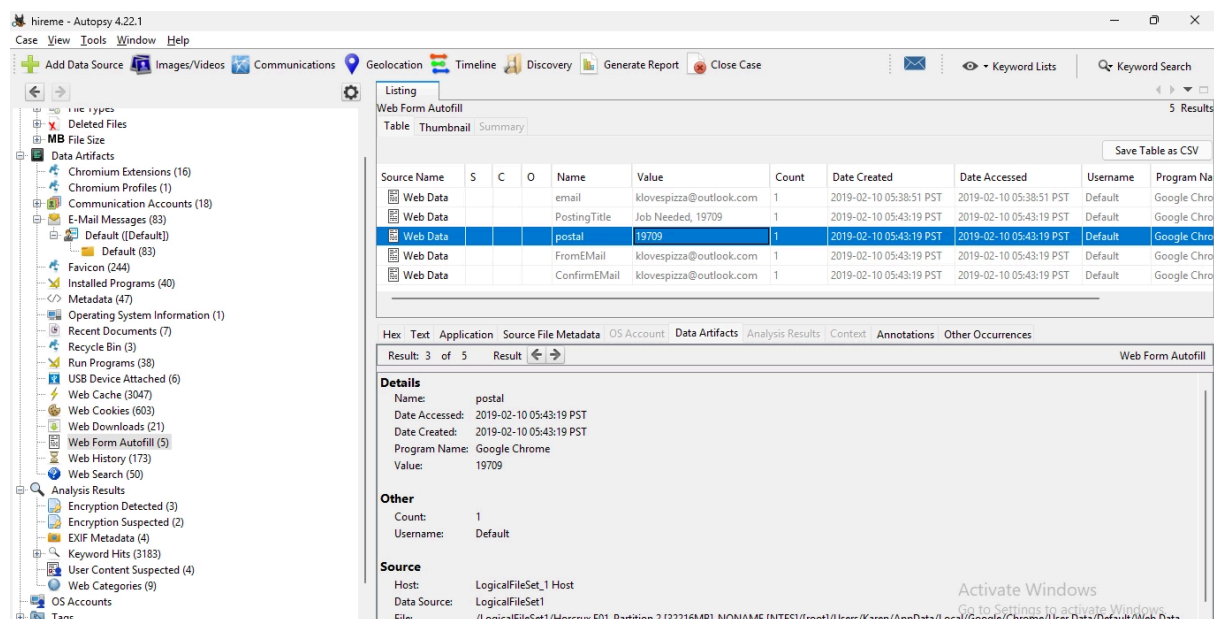
Based on the hint provided, we focus on analyzing **web browser data**.

A quick search suggests that **ZIP/postal codes** used by browsers are typically associated with user location data.

Location of Artifact

This information can be found in web browser autofill data:

- **Web form autofill / Web Data → Postal**



Q6

Solved : 1327

What are the initials of the person who contacted the admin user from TAAUSAI?

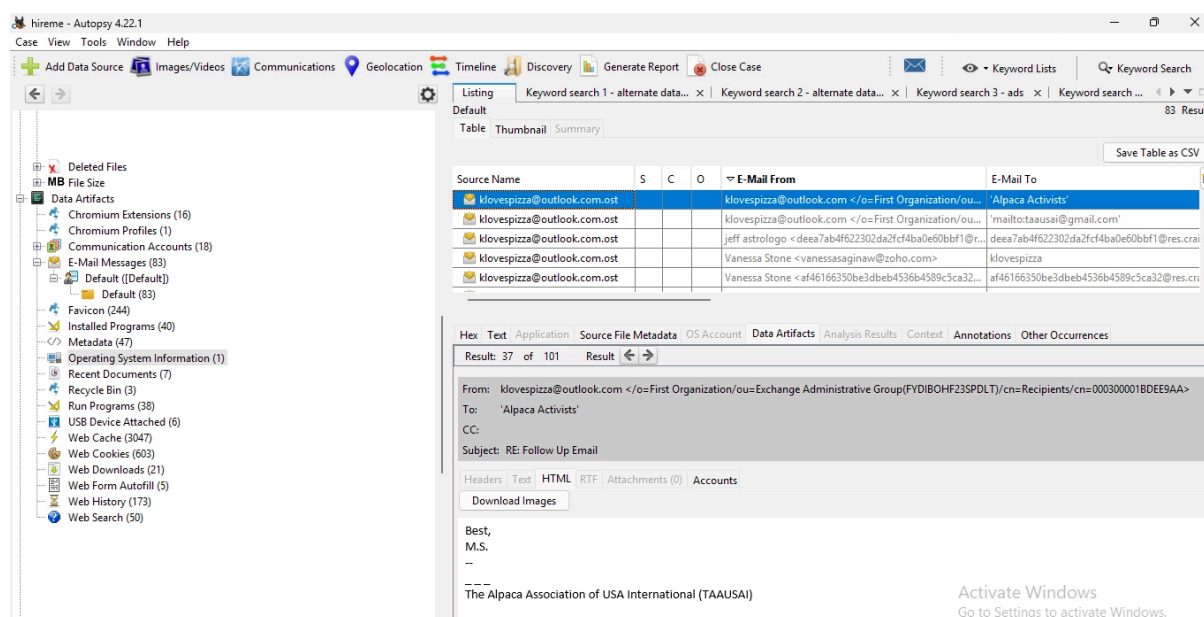
flag: ms

process:

We then proceeded to the **Messages** section and examined the available conversations.

During the analysis, we identified a message from **The Alpaca Association of USA International (TAAUSAI)**.

The sender of the message was **Michael Scotch**, whose initials can be noted as **MS**.



Q7

Solved : 1328

How much money was TAAUSAI willing to pay upfront?

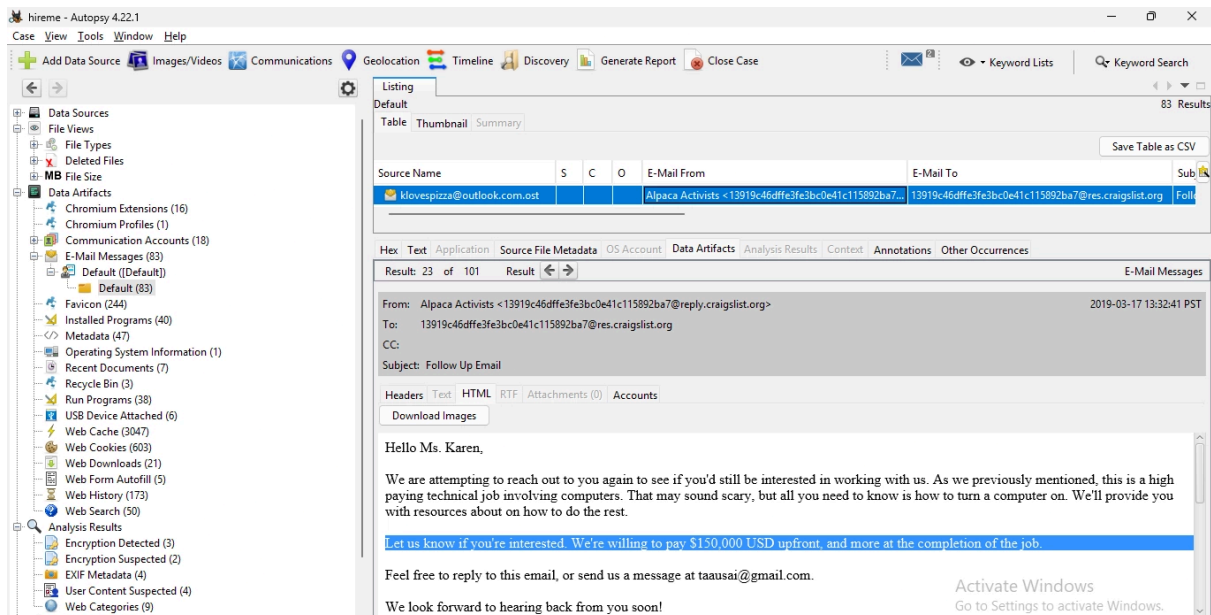
flag: 150000

process:

As we continued scrolling through the conversations between the user and the organization, we discovered another message:

"Let us know if you're interested. We're willing to pay \$150,000 USD upfront, and more at the completion of the job."

This message was sent by **M.S. (Michael Scotch)** from **The Alpaca Association of USA International (TAAUSAI)**.



Q8

Solved : 1292

What country is the admin user meeting the hacker group in?

flag: Egypt

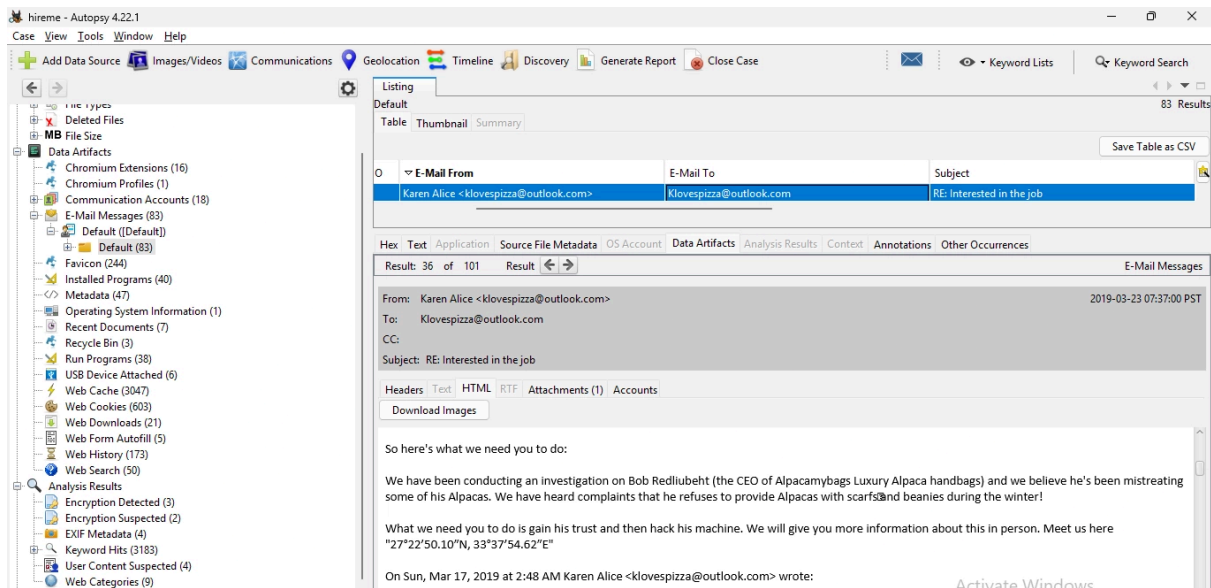
process:

In this conversation between **Alpaca Activists (taausai@gmail.com)** and **Karen Alice (klovespizza@outlook.com)**, it was revealed that they have been conducting an investigation on a CEO.

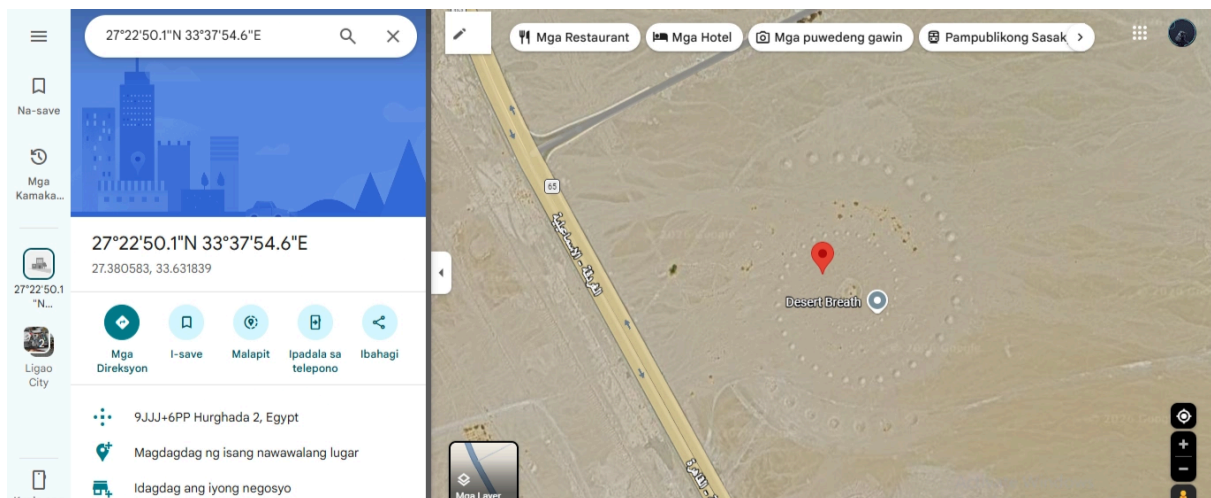
They instructed Karen Alice to gain the CEO's trust and then proceed to hack his machine. They also mentioned that further information about the target would be provided later.

Additionally, the user was instructed to meet at the following coordinates:

27°22'50.10"N, 33°37'54.62"E



This location was searched using Google Maps, and it was identified as being in **Egypt**.



Q9

Solved : 1351

What is the machine's timezone? (Use the three-letter abbreviation)

flag: UTC

process:

Let us go back to **Registry Explorer** and load the **SYSTEM** hive again.

We then used the **Quick Search** function and searched for **"timezone"**.

Registry Explorer v2.1.0

FileToolsOptionsBookmarks (35/0)ViewHelp

Registry hives (2)Available bookmarks (37/0)

timezone

Find

Key name	# values	# subkeys	Last write timestamp
C:\Users\joki\Downloads\62-...	=	=	=
ROOT	0	15	2019-03-22 23:21:35
ControlSet001	0	6	2017-09-29 13:47:03
Control	12	109	2019-03-22 23:58:30
TimeZoneInformation	10	0	2019-03-17 22:03:25
Services	0	635	2019-03-22 23:58:30
SystemEventsBroker	11	3	2017-09-29 13:47:44
Parameters	2	1	2017-09-29 13:47:03
EventPolicyTable	0	33	2017-09-29 13:47:03
SebTimeZoneCh...	1	0	2017-09-29 13:47:03

ValuesTimeZoneInformation

Drag a column header here to group by that column

Value Name	Value Data	Value Data Raw
Bias	0	0
DaylightBias	0	0
DaylightName	@tzres.dll,-931	@tzres.dll,-931
DaylightStart	Month 0, week of month 0, day of week 0, Hours:Minutes:Seconds:Milliseconds 0:0:0:0	00-00-00-00-00-00-00-00-00-00-00-00-00-00-00
StandardBias	0	0
StandardName	@tzres.dll,-932	@tzres.dll,-932
StandardStart	Month 0, week of month 0, day of week 0, Hours:Minutes:Seconds:Milliseconds 0:0:0:0	00-00-00-00-00-00-00-00-00-00-00-00-00-00-00
TimeZoneKeyName	UTC	UTC
ActiveTimeBias	0	0

Total rows: 9Export?

Type viewerBinary viewer

Value nameBias

Value typeRegDword

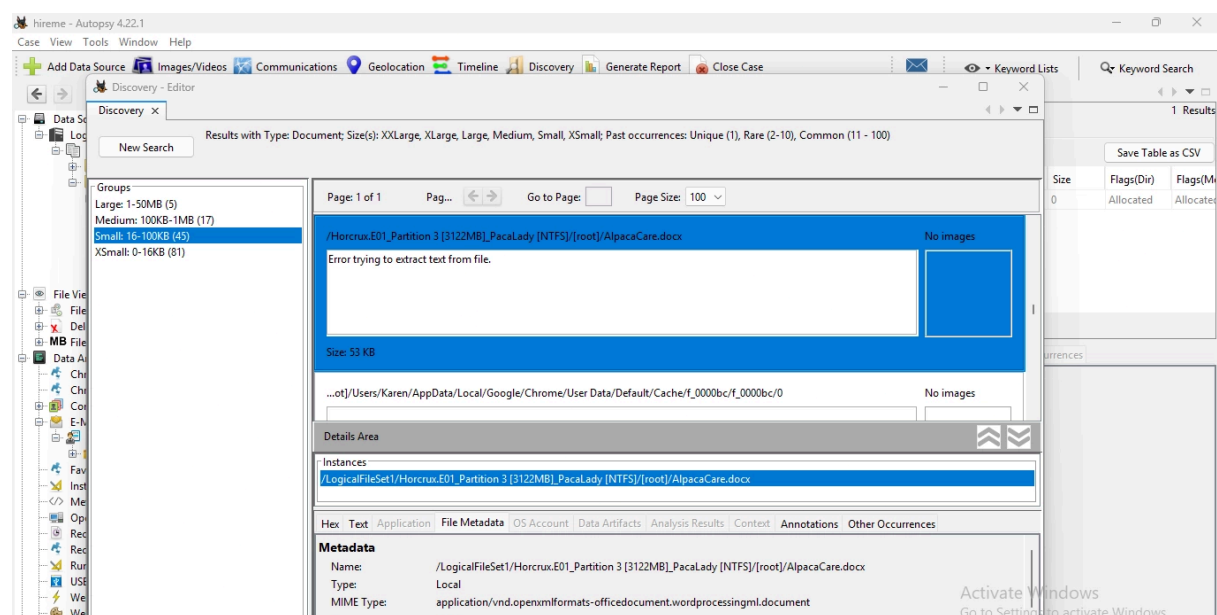
Value0

Solved : 1235

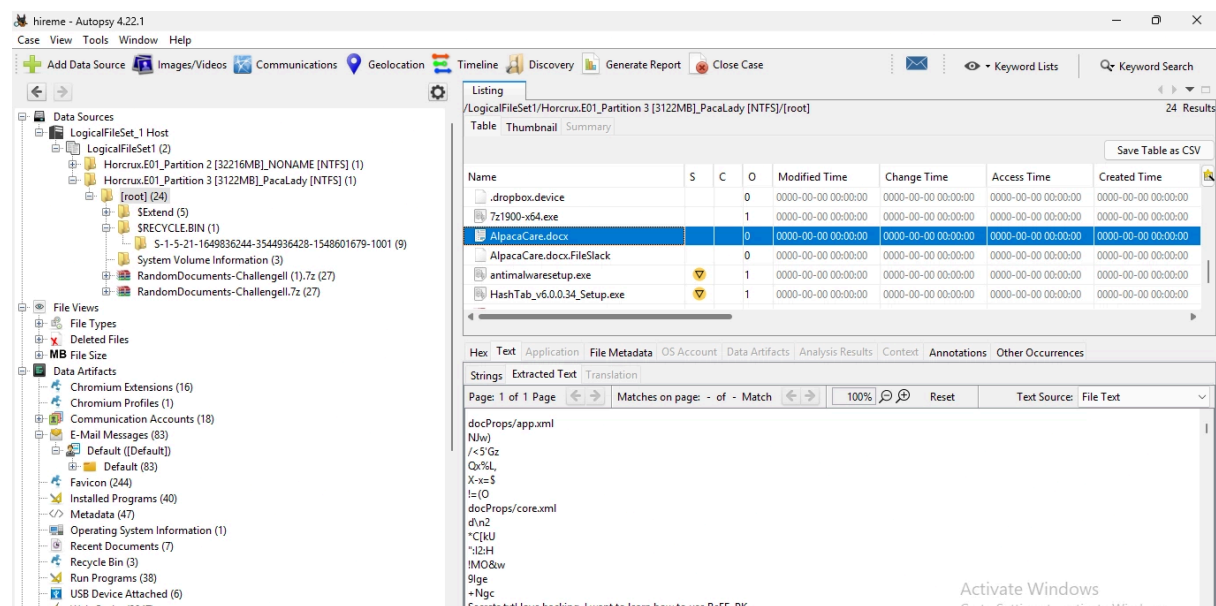
flag: 2019-03-17 21:52

process:

Let's use the **Discovery** tool in Autopsy to locate the specified file.



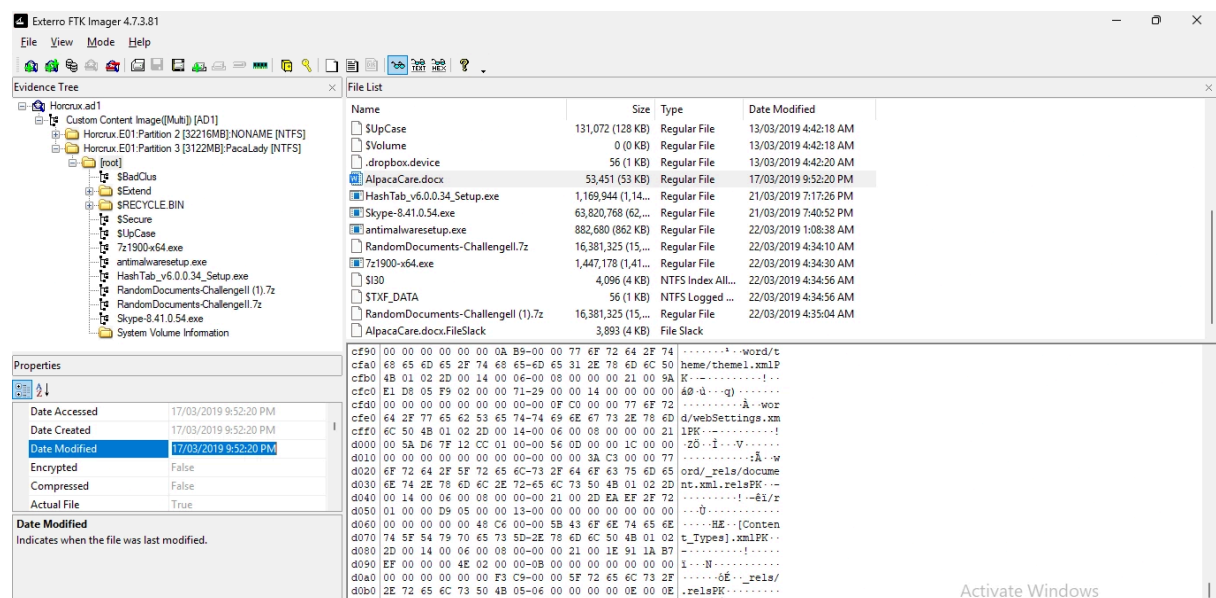
We navigated to the **Documents** section, selected **All File Sizes**, and performed a search for the target file.



After locating it, we identified that the file is stored in the **Partition 3 root directory**.

However, within Autopsy, no timestamp information was clearly specified. To verify the metadata, we proceeded to **FTK Imager** and located the same file there.

In FTK Imager, we were able to view the **last accessed time** of the file.



The timestamp was recorded in **12-hour format** as:

This was then converted into **24-hour format** as:

- 17/03/2019 21:52:20 PM

Q11

Solved : 1303

There was a second partition on the drive. What is the letter assigned to it?

flag: A

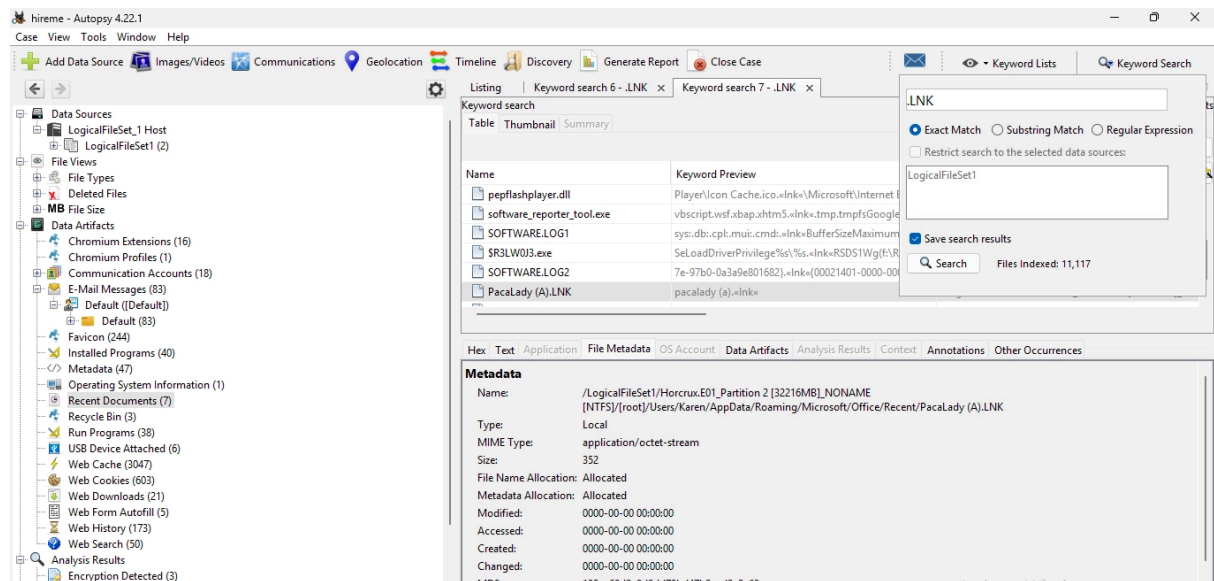
process:

In the **Recent Documents** section, a **.LNK** file was identified at the following path:

/LogicalFileSet1/Horcrux.E01_Partition 2 [32216MB]_NONAME
[NTFS]/[root]/Users/Karen/AppData/Roaming/Microsoft/Office/Recent/PacaLady (A).LNK

This shortcut file indicates that the file **"PacaLady"** from **Partition 2** was assigned the drive letter **A:**.

A quick search on `.LNK` file artifacts confirms that Windows shortcut files often store metadata about the original file location, including the assigned drive letter. In this case, it verifies that **Partition 2 was mapped to drive letter A.**



Q12

Solved : 1249

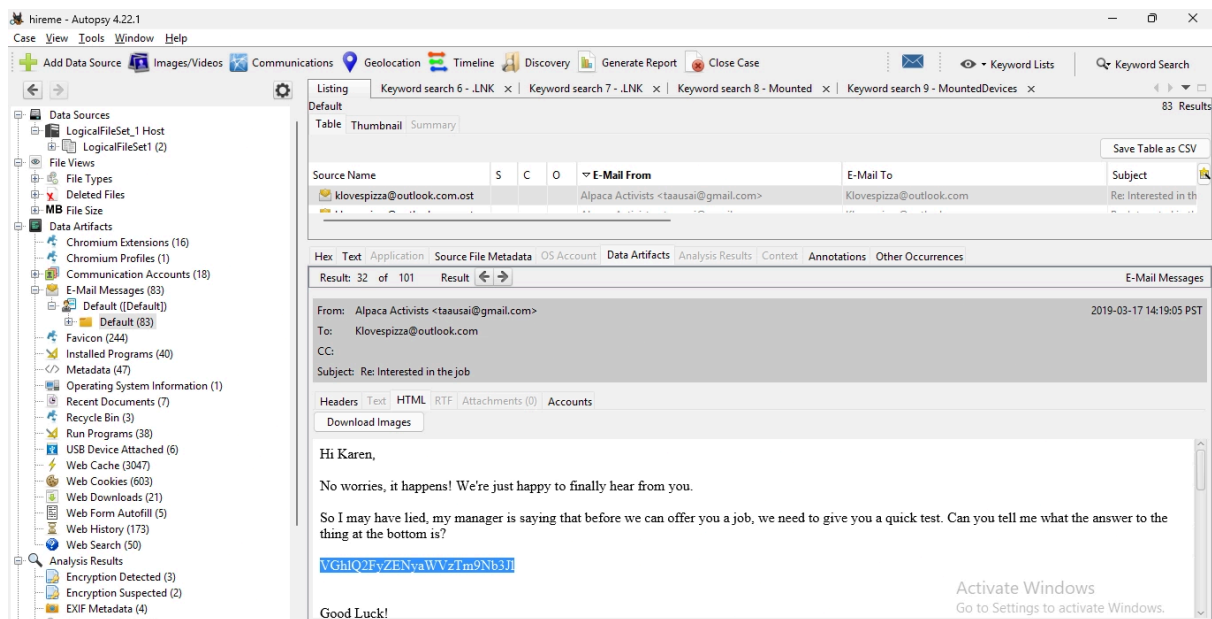
What is the answer to the question Company's manager asked Karen?

flag: TheCardCriesNoMore

process:

From the message sent by **Alpaca Activist** to **klovespizza**, the following was included:

"So I may have lied, my manager is saying that before we can offer you a job, we need to give you a quick test. Can you tell me what the answer to the thing at the bottom is?"

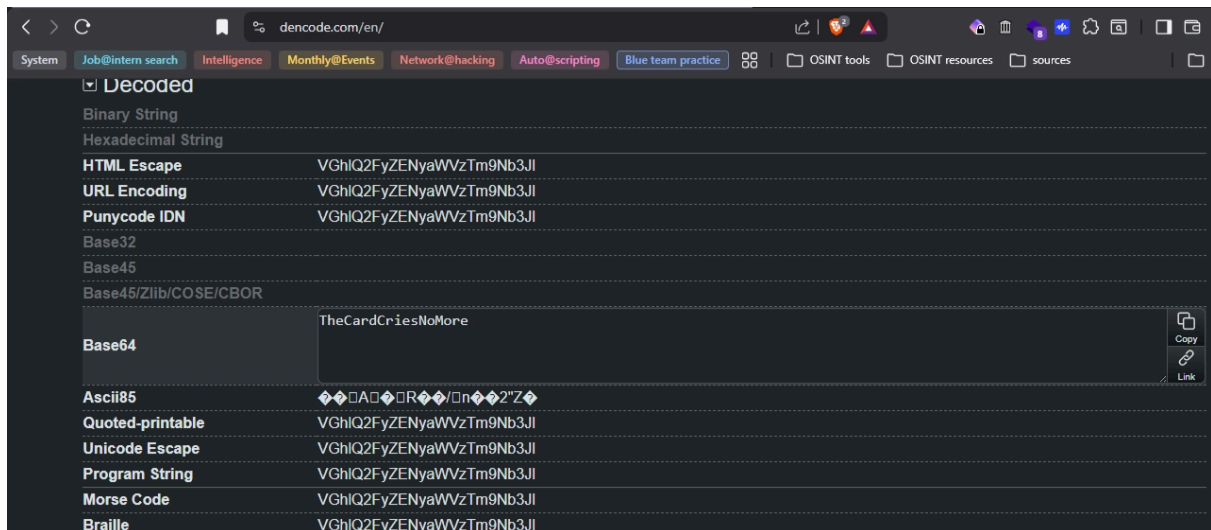


The string provided for decoding was:

VGhIQ2FyZENyaWVzTm9Nb3Jl

When decoded from Base64, it resolves to:

TheCardCriesNoMore



Q13

Solved : 1251

What is the job position offered to Karen? (3 words, 2 spaces in between)

flag: cyber security analyst

process:

Between the conversation:

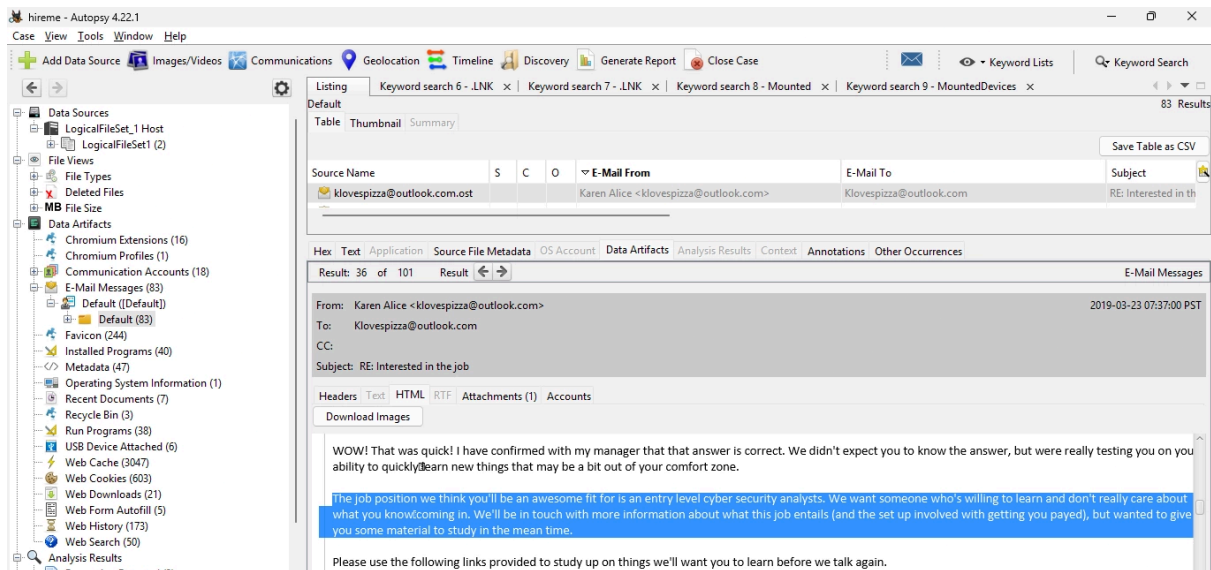
From: Alpaca Activists (taausai@gmail.com)

To: Karen Alice (klovespizza@outlook.com)

Subject: Re: Interested in the job

We identified the following message from **Alpaca Activists**:

"The job position we think you'll be an awesome fit for is an entry level cyber security analyst. We want someone who's willing to learn and don't really care about what you know coming in. We'll be in touch with more information about what this job entails (and the setup involved with getting you paid), but wanted to give you some material to study in the meantime."



This message outlines that the intended role being offered is an **entry-level cybersecurity analyst position**, with an emphasis on willingness to learn rather than prior experience.

Q14

Solved : 1222

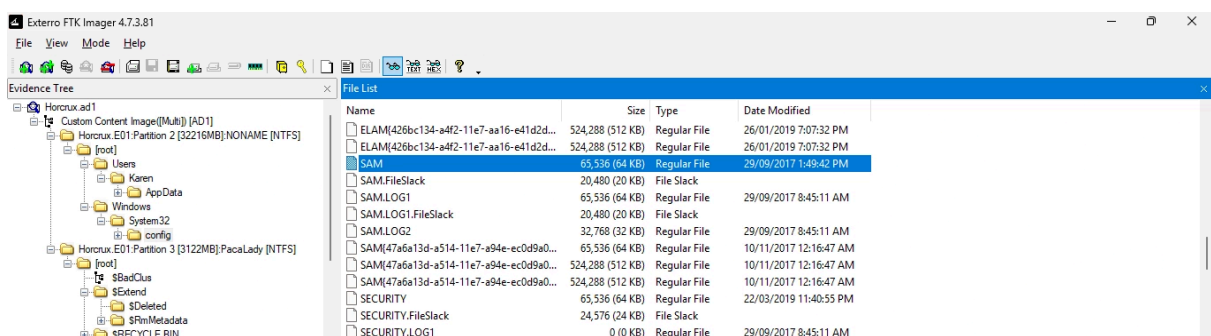
When was the admin user password last changed?

flag: 03/21/2019 19:13:09

process:

A quick search indicated that records related to **local user password changes** are primarily stored in the **Security Accounts Manager (SAM) database**, located at:

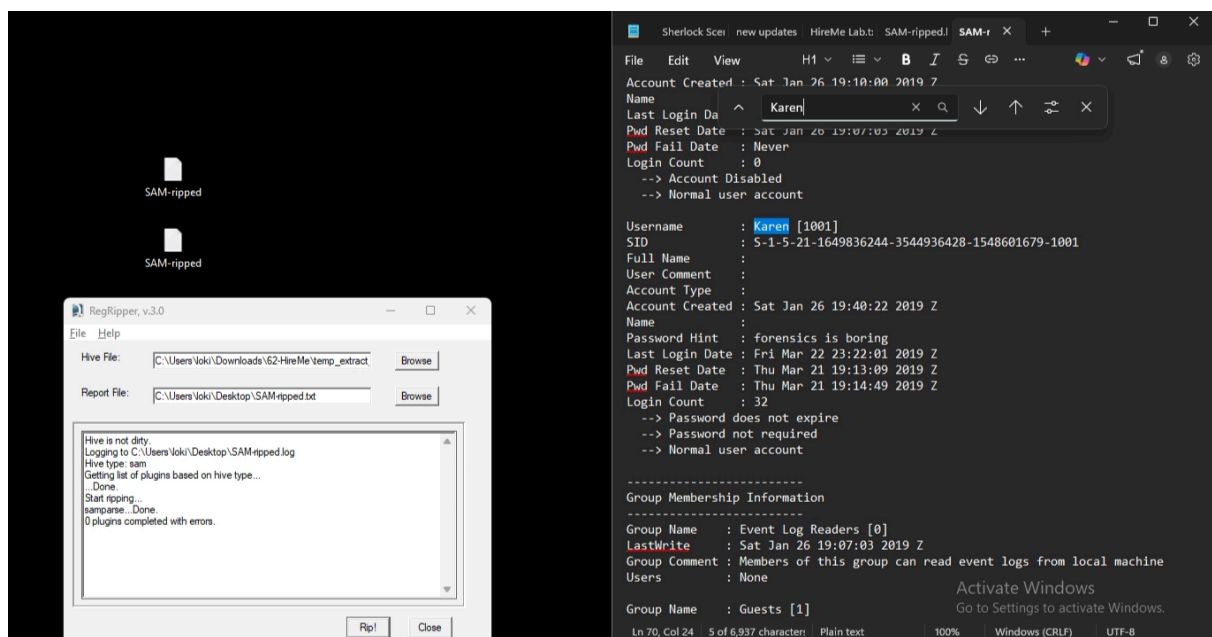
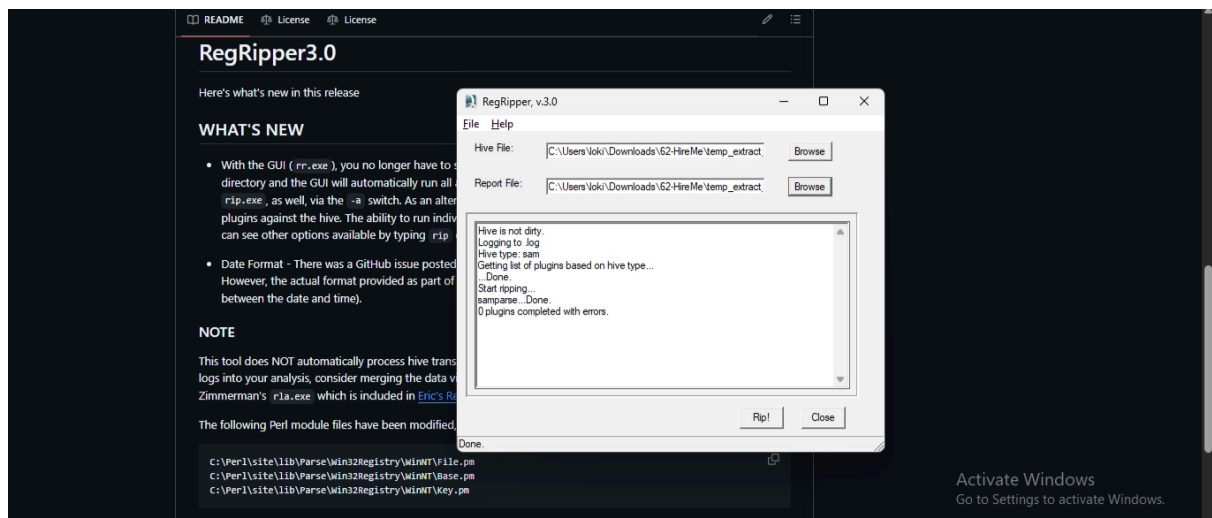
C:\Windows\System32\config\SAM



We then extracted the **SAM file** and attempted to analyze it using **Registry Explorer** by loading it as a hive. However, no relevant results were found.

Further research revealed that the SAM database is typically **encrypted**, making direct analysis more difficult. As a result, we identified a more suitable tool for this purpose: **RegRipper**.

We launched **RegRipper**, loaded the extracted SAM file, and performed a **RIP operation** to generate a report in **.txt** format.



After opening the report, we searched for **"Karen"** and found the relevant entry:

- **Password Reset Date:** Thu Mar 21 19:13:09 2019 Z

Q15

Solved : 1266

What version of Chrome is installed on the machine?

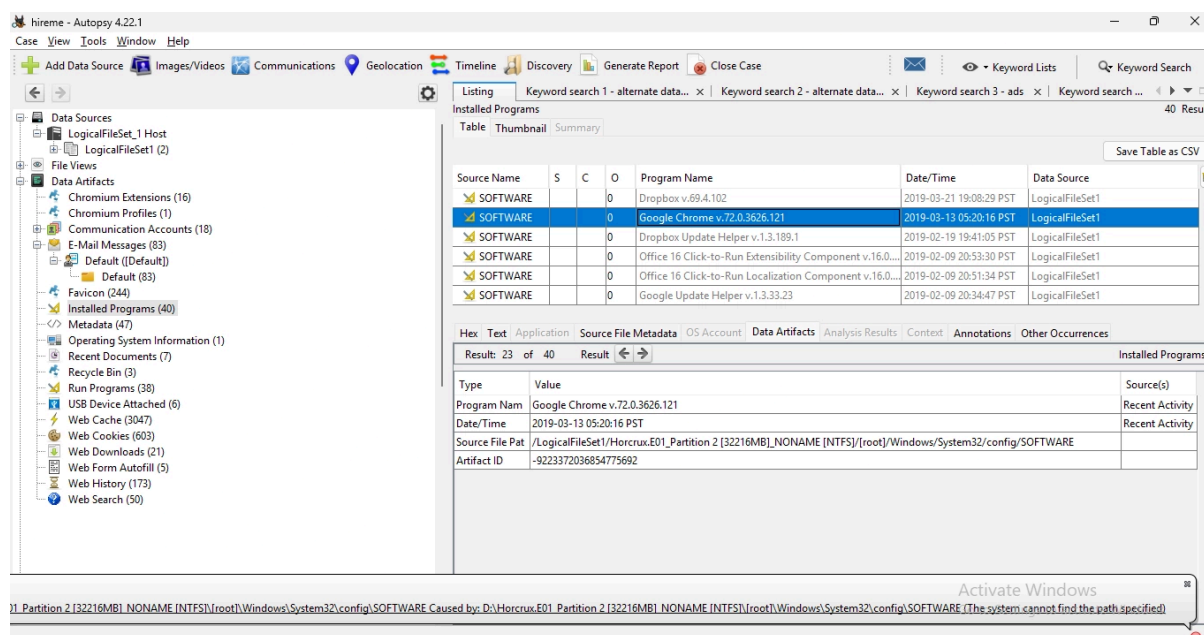
flag: 72.0.3626.121

process:

In **Autopsy**, we proceeded to the **Installed Programs** section and identified the software installed on the system.

Among the listed applications, we found **Google Chrome**, which was installed on the system. The installed version was:

- **Google Chrome v72.0.3626.121**



Q16

Solved : 1214

What is the URL used to download Skype?

flag:<https://download.skype.com/s4l/download/win/Skype-8.41.0.54.exe>

process:

As indicated by the hint, the URL is stored within the **Alternate Data Stream (ADS)** of the **Skype executable**.

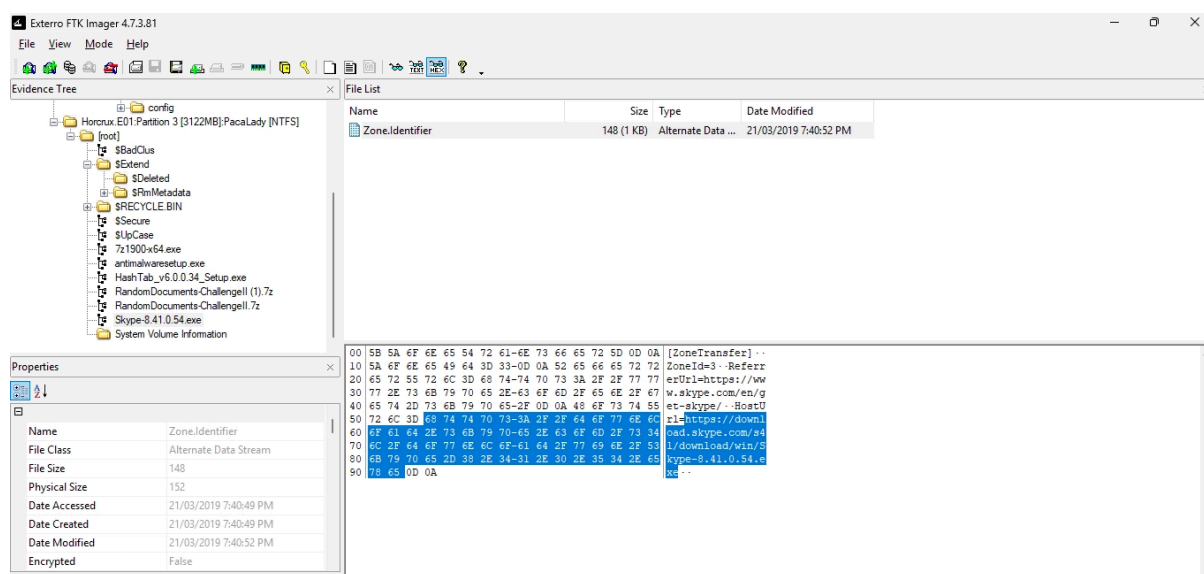
An **Alternate Data Stream (ADS)** is a feature of the NTFS file system that allows files to contain hidden data streams separate from the main file content. These streams are not visible through standard file browsing methods and are often

used for metadata such as the **Zone.Identifier**, which tracks the origin of downloaded files.

Using **FTK Imager**, we navigated to the `skype.exe` file and inspected its Alternate Data Streams. There, we identified the **Zone.Identifier** stream containing the required information.

The extracted URL is:

<https://download.skype.com/s4l/download/win/Skype-8.41.0.54.exe>



Q17

Solved : 1198

What is the domain name of the website Karen browsed on Alpaca care that the file AlpacaCare.docx is based on?

flag: palominoalpaca.com

process:

We first checked the **Web Data** section in Autopsy, but no relevant information was found.

We then performed a quick search and discovered that we could extract the file for deeper analysis. The file was exported and analyzed on a Linux.

File Extraction

We used the following command to extract the contents:


```
strings document.xml | grep "http://"
```

Result

This led us to the following extracted URL:

HYPERLINK "<http://palominoalpaca.com/>"